Brainstorming Document

Ideation Phase: Technology choices and user journey

Brainstorming Document

Project: FitBuddy — AI-Powered Personal Fitness Coach

Brainstorming Session Overview

Date: March 2025
Focus Area: AI-Powered Fitness Application using Google Gemini API
Goal: Identify the best approach to build a scalable, AI-driven fitness coach app

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Mind Map: Core Ideas

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■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ▼ ▼ ▼ ▼ [User Input] [AI Engine]
[Data Storage] [Frontend UI] ■ ■ ■ ■ Name, Age, Gemini Flash SQLite DB Dark Mode Weight, Goal, LLM
Prompts Users Table Animated Intensity JSON Output Plans Table Responsive Nutrition Cards/Tabs
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Feature Brainstorming

Core Features (Must-Have)

1. **Personalized Workout Plan Generation** — Using Gemini AI with structured prompts
2. **7-Day Plan Structure** — Day-by-day breakdown with warmup, exercises, cooldown
3. **Feedback-Based Plan Update** — Users describe changes, AI regenerates accordingly
4. **Nutrition & Recovery Tips** — Tailored dietary advice based on goal

5. **User Profile Storage** — Save user data to SQLite for persistence

Nice-to-Have Features (Explored)

- History tracking per user
- Exercise video links
- Calorie calculator integration
- Weekly progress tracking
- Push notifications

Discarded Features (Too Complex)

- Real-time chat with AI coach
- Wearable device integration
- Payment/subscription model
- Social sharing features

Technology Brainstorming

Layer	Options Considered	Selected	Reason
Backend	Flask, FastAPI, Django	**FastAPI**	Async, fast, auto-docs, Pydantic
AI	OpenAI, Cohere, Gemini	**Gemini 2.0 Flash**	Free tier, powerful, fast
Database	PostgreSQL, MongoDB, SQLite	**SQLite**	Simple, file-based, no setup
Frontend	React, Vue, Vanilla HTML	**Vanilla HTML/CSS/JS**	Lightweight, no build step
Styling	Bootstrap, Tailwind, Custom	**Custom CSS**	Full control, glassmorphism
Deployment	AWS, Heroku, Vercel	**Local + Docker**	Easy for submission

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User Journey Brainstorm

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User Opens App ■ ▼ Fill Profile Form (Name, Age, Weight, Goal, Intensity) ■ ▼ Click "Generate My
7-Day Plan" ■ ▼ AI Generates Personalized Plan → Displayed with Cards ■ ▼ User Reviews → Gives
Feedback → Click "Update Plan" ■ ▼ AI Refines Plan → Updated Plan Shown ■ ▼ User Checks
"Nutrition Tips" Tab → Enters Goal → Gets Tips
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Prompt Engineering Ideas

Brainstormed Prompt Approaches:

1. **Simple prompt**: "Create a workout plan for {name}" — Too generic ■
2. **Structured prompt with JSON output**: Gives the AI a schema to fill — Best ■
3. **Chain-of-thought prompt**: Ask AI to reason first, then plan — Too verbose ■
4. **Role-based prompt**: "You are an expert fitness coach..." — Great for tone ■

Final Approach:

- Role-based system prompt + user profile data + required JSON schema
- Response parsing with markdown code fence stripping
- Temperature controlled for consistency

Risk Brainstorm

Risk	Likelihood	Mitigation
Gemini API rate limits	Medium	Cache plans, add error handling
Invalid JSON from AI	Medium	Try-except + JSON validation
SQLite concurrency	Low	Single-user setup for demo
API key exposure	High	Environment variables
UI not responsive	Low	CSS grid + media queries

Conclusion

The brainstorming session led to the following key decisions:
- **FastAPI + Gemini + SQLite** as the core tech stack
- **Three core scenarios** as the main features
- **Glassmorphism dark UI** for maximum visual impact
- **Structured JSON prompts** for reliable AI output